

C2 Assignment : Coordinate Geometry Vector 3D

SINGLE CORRECT TYPE QUESTIONS

C02/1/001. If two real number x and y satisfy $(x + 5)^2 + (y - 10)^2 = 196$, then the minimum possible value of $x^2 + 2x + y^2 - 4y$ is :

- (A) $271 - 112\sqrt{5}$ (B) $14 - 4\sqrt{5}$
(C) $276 - 112\sqrt{5}$ (D) $9 - 4\sqrt{5}$

Ans. A

Sol. Denote by C the circle $(x + 5)^2 + (y - 10)^2 = 196$.

If $x^2 + 2x + y^2 - 4y = r$, then $(x + 1)^2 + (y - 2)^2 = r + 5$ and the point (x, y) belongs to the circle of radius $\sqrt{r + 5}$ and with center $(-1, 2)$. The point $(-1, 2)$ is in the interior of the circle C. The smallest possible value of r taken by points (x, y) corresponds to the radius of a circle with center $(-1, 2)$ for which it touches the circle C. Consider the line the passes through centers of circles $(-1, 2)$ and $(-5, 10)$:

$$\begin{cases} x = -5 + 4t \\ y = 10 - 8t \end{cases}$$

Where $t \geq 0$. The point $(-5 + 4t, 10 - 8t)$ belongs to the circle C when

$$16t^2 + 64t^2 = 196, t = \sqrt{\frac{49}{20}} = \frac{7}{2\sqrt{5}}.$$

Hence, the minimal possible radius r satisfies

$$r + 5 = (-4 + 4t)^2 + (8 - 8t)^2 = 80(1 - t)^2 = 80\left(1 - \frac{7}{\sqrt{5}} + \frac{49}{20}\right)$$

$$= 80 - 112\sqrt{5} + 196 = 276 - 112\sqrt{5}$$

$$\text{Hence } r = 271 - 112\sqrt{5}$$

C02/1/002. Let $A \equiv (0, 4)$ and $B \equiv (3, 8)$ be two points such that $\angle AXB$ is maximum where X is a point on x -axis.

If x -co-ordinate of X is $a\sqrt{2} - b$, then $(a^4 + b^4)$ is :

- (A) 106 (B) -602 (C) 17 (D) 706

Ans. D

Sol.

C04/1/003. If a circle of radius 5 is concentric with ellipse $3x^2 + 7y^2 = 105$, then m is slope of the common tangent to circle and ellipse. If normal at $P(\theta)$ to ellipse $5x^2 + 14y^2 = 70$ intersects it again at $Q(2\theta)$, then $m^2 - 3 \cos \theta$ is equal to :

- (A) 1 (B) 2 (C) 3 (D) 4

Ans. C

Sol.

V01/1/004. Given vectors $\vec{a} = \sin \frac{\pi}{6} \hat{i} + \sin \frac{\pi}{3} \hat{j}$; $\vec{b} = \sin \frac{\pi}{3} \hat{j} + \sin \frac{\pi}{6} \hat{k}$ and $\vec{c} = \hat{i} + 2\hat{j} + 3\hat{k}$. The volume of parallelepiped with three conterminous edges as $\vec{u} = (\vec{a} \cdot \vec{a})\vec{a} + (\vec{a} \cdot \vec{b})\vec{b} + (\vec{a} \cdot \vec{c})\vec{c}$, $\vec{v} = (\vec{a} \cdot \vec{b})\vec{a} + (\vec{b} \cdot \vec{b})\vec{b} + (\vec{b} \cdot \vec{c})\vec{c}$, $\vec{w} = (\vec{a} \cdot \vec{c})\vec{a} + (\vec{b} \cdot \vec{c})\vec{b} + (\vec{c} \cdot \vec{c})\vec{c}$ equals:

(A) $\left(2 \cos \frac{\pi}{4} - \sin \frac{\pi}{3}\right)^3$

(B) $\left(2 \cos \frac{\pi}{3} - \sin \frac{\pi}{3}\right)^3$

(C) $\left(2 \cos \frac{\pi}{6} - \sin \frac{\pi}{6}\right)^3$

(D) $\left(2 \sin \frac{\pi}{4} - \cos \frac{\pi}{3}\right)^3$

Ans. C

Sol.

$$[\vec{u} \vec{v} \vec{w}] = \begin{vmatrix} \vec{a} \cdot \vec{a} & \vec{a} \cdot \vec{b} & \vec{a} \cdot \vec{c} \\ \vec{b} \cdot \vec{a} & \vec{b} \cdot \vec{b} & \vec{b} \cdot \vec{c} \\ \vec{c} \cdot \vec{a} & \vec{c} \cdot \vec{b} & \vec{c} \cdot \vec{c} \end{vmatrix} [\vec{a} \vec{b} \vec{c}]$$

$$= [\vec{a} \vec{b} \vec{c}]^3$$

$$[\vec{a} \vec{b} \vec{c}] = \begin{vmatrix} \frac{1}{2} & \frac{\sqrt{3}}{2} & 0 \\ 0 & \frac{\sqrt{3}}{2} & \frac{1}{2} \\ 0 & 2 & 3 \end{vmatrix} = \left(\sqrt{3} - \frac{1}{2}\right)$$

V01/1/005. A particle move over a path, whose position vector at any time t , is given by $\vec{r}(t) = |t|\hat{i} + |\sin \pi t|\hat{j} + \sqrt[3]{t-2}\hat{k}$. A path $\vec{r}(t)$ is called differentiable at time t if its all components are differentiable with respect to 't' at that point of time. So $\vec{r}(t)$ is differentiable at $t =$:

(A) 3

(B) 1

(C) 2

(D) 3.5

Ans. D

Sol. Function $|t|$ is not differentiable only at $t=0$, whereas function $|\sin \pi t|$ is differentiable every where except at all integral points. Also, function $\sqrt[3]{t-2}$ is not differentiable only at $t=2$. So, $\vec{r}(t)$ is differentiable every where except at all integral points.

ONE OR MORE THAN ONE CORRECT TYPE QUESTIONS

C01/2/006. Minimum value of $\sqrt{2x^2 + 2x + 1} + \sqrt{2x^2 - 10x + 13}$ is $\sqrt{\alpha}$, then :

- (A) α is an even number (B) sum of digits of α is 2
(C) number of prime factors of α is 2 (D) number of prime factors of α is 3

Ans. ABC

Sol. $\sqrt{(\alpha+1)^2 + (\alpha+2-2)^2} + \sqrt{(\alpha-2)^2 + (\alpha+2-5)^2}$

Use concept of Optimization for PA + PB to be minimum using line equation as $y = x + 2$ and P($\alpha, \alpha + 2$) is any point on this line and A $\equiv (2, 5)$ and B $\equiv (-1, 2)$

C01/2/007. If A_i is the area bounded by $|x - a_i| + |y| = b_i$, $i \in \mathbb{N}$, where $a_{i+1} = a_i + \frac{3}{2}b_i$ and $b_{i+1} = \frac{b_i}{2}$, $a_1 = 0$, $b_1 = 32$, then :

- (A) $A_3 = 128$ (B) $A_3 = 256$
(C) $\lim_{n \rightarrow \infty} \sum_{i=1}^n A_i = \frac{8}{3}(32)^2$ (D) $\lim_{n \rightarrow \infty} \sum_{i=1}^n A_i = \frac{4}{3}(16)^2$

Ans. AC

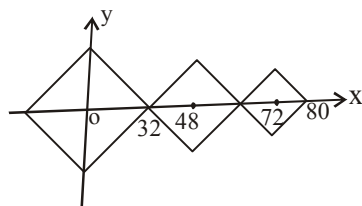
Sol. $a_2 = a_1 + \frac{3}{2}b_1 = 48, b_2 = \frac{b_1}{2} = 16$

$a_3 = 48 + \frac{3}{2} \times 16 = 72, b_3 = \frac{16}{2} = 8$

So, the three loops from $i = 1$ to $i = 3$ are alike Now, area of i^{th} loop (square) = $\frac{1}{2}(\text{diagonal})^2$

$A_i = \frac{1}{2}(2b_i)^2 = 2b_i^2$

$\therefore \frac{A_{i+1}}{A_i} = \frac{2(b_{i+1})^2}{2(b_i)^2} = \frac{1}{4}$



So, the areas form a G.P. series

$\therefore$ sum of the G.P. up to infinite term is $\frac{A_i}{1-r} = \frac{2(32)^2}{1-\frac{1}{4}} = \frac{8}{3}(32)^2$ sq. units.

- C01/2/008. Let $x + y = 2$ and $2x + y = 5$ are internal angle bisectors of angles B and C of ΔABC respectively. If $A(3, 4)$, then which of the following options is/are correct :
- (A) Image of A about internal bisector of angle B is $(-2, -1)$.
- (B) If r denotes inradius of ΔABC , then $2r = 3\sqrt{10}$
- (C) Equation of BC is $y = 3x + 5$
- (D) Image of A about internal bisector of angle C is $(-1, -2)$.

Ans. ABC

Sol.

- C01/2/009. Let $L \equiv ax + by + c = 0$, $M \equiv bx + cy + a = 0$, $N \equiv cx + ay + b = 0$.
If $\lambda = a + b + c$, $\mu = a^2 + b^2 + c^2$, $v = ab + bc + ca$; $a, b, c \in \mathbb{R}$, then which of the following options is/are correct :
- (A) If $\lambda = 0$, $\mu \neq v$, then the lines L, M, N always pass through a fixed point.
- (B) If $\lambda = 0$, $\mu = v$, then each of L, M, N are true for all points in xy-plane.
- (C) If $\lambda \neq 0$, $\mu \neq v$, then lines L, M, N form a triangle.
- (D) If $\lambda \neq 0$, $\mu = v$, then graph of L, M, N are same.

Ans. ABCD

Sol.

- C01/2/010. Let M and N be line mirrors. A ray of light along L_1 strikes M and reflects along line L_2 . After second reflection from N, the ray moves along line L_3 and after third reflection from M, it moves along line L_4 .
If $L_1 \equiv x - 3y = 1$, $M \equiv y = 0$ and $N \equiv x = 4$, then which of the following options is/are correct :
- (A) Equation of line L_4 is $x + 3y - 5 = 0$
- (B) Area of the figure so formed is 12 square units.
- (C) Equation of line L_3 is $x - 3y = 7$
- (D) Equation of line L_2 is $x + 3y = 1$

Ans. CD

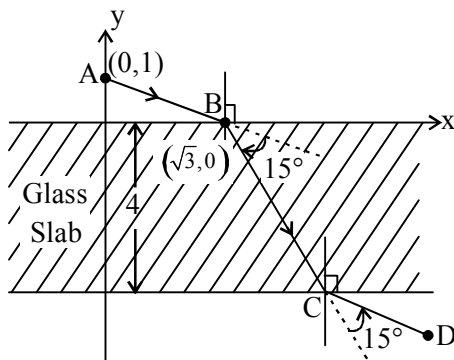
Sol.

- C01/2/011. Let L_i ; $i = 1, 2, 3, \dots, 7$ be lines such that $L_1 \equiv x + y = 1$, $L_2 \equiv x + y = 3$, $L_3 \equiv 2x - y = 5$, $L_4 \equiv 3x + y = 5$,
Line L_5 passes through origin, and meets L_1 and L_2 at P and Q respectively. Through P and Q, lines L_6 and L_7 are drawn parallel to L_3 and L_4 respectively intersecting at point R. Which of the following option(s) can be point R:
- (A) (4, 3) (B) (10, 4) (C) (19, 7) (D) (25, 10)

Ans. AD

Sol.

- C01/2/012. A ray of light from A to glass slab of thickness 4 units is incident at point B(as shown in figure). After refraction, it turns by 15° towards the normal and moves along BC. After second refraction it turns by 15° away from the normal, and moves along CD.



If $CD = AB$, then which of the following options is / are correct :

- (A) Equation of BC is $x + y = \sqrt{3}$ (B) Coordinates of C are $(2 + \sqrt{3}, -4)$
 (C) Equation of DC is $\sqrt{3}y + x = 2 - 3\sqrt{3}$ (D) Coordinates of D are $(2\sqrt{3} + 4, -5)$

Ans. AD

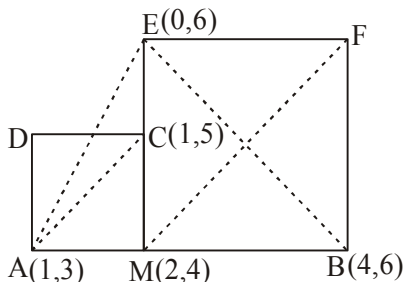
Sol.

- C02/2/013. A point M divides A and B in the ratio 1 : 2 where A and B are diametrically opposite ends of a circle $x^2 + y^2 - 5x - 9y + 22 = 0$. Square AMCD and BMEF on the length AM and MB are constructed on the same side of line AB. If co-ordinates of A is (1, 3) then the orthocenter of $\triangle ABE$ is :

- (A) (1, 6) (B) (1, 5) (C) (3, -3) (D) (4, 6)

Ans. BC

Sol. As shown in the figure since C (1, 5) is the orthocenter of triangle $\triangle AEB$, similarly for the other side, the coordinate is C (3, 3)



- C02/2/014. Circle $x^2 + y^2 - 8x - 6y + 16 = 0$ and line $y = mx + c$ intersect at two distinct points A and B, such that circle with AB as diameter always passes through origin. If $x^2 + y^2 + 2gx + 2fy + k = 0$ represents the locus of center of circle with AB as diameter, then :

- (A) $c^2 + (4m - 3)c + 8(m^2 + 1) = 0$ (B) $k + g = 6$

(C) $4c^2 + (4m - 3)c + 8(m^2 + 1) = 0$

(D) $k - 2f = 11$

Ans. ABD

Sol.

C02/2/015. Let $x^2 + y^2 + 2gx + 2fy + c = 0$ represent the equation of circle passing through points lying in $A \cup B$, given that it exists and $|A| = 2 = |B|$, where $|X|$ denotes the cardinal number of set X ,
 $A = \{(x, y) \in \mathbb{R}^2 | x^2 + y^2 = 4, x + 2y = a\}$, $B = \{(x, y) \in \mathbb{R}^2 | x^2 + y^2 - 4x - 2y + 1 = 0, 12x - 6y = 41\}$. Then which of the following options is / are correct :

(A) $a + 2 = 0$

(B) $5g - 5f = 4$

(C) $c = -36$

(D) $a - 5c = 34$

Ans. ABD

Sol.

C02/2/016. Circle C_1 touches X-axis at a distance of 3 units from origin, and has an intercept of length $2\sqrt{7}$ on Y-axis. Circle C_2 passes through point $(0, 1)$ and is orthogonal to $C_3 \equiv (x - 1)^2 + y^2 = 16$ and $C_4 \equiv x^2 + y^2 = 1$. Then which of the following options is / are correct :

(A) Possible equation of C_1 is $x^2 + y^2 - 6x + 8y + 9 = 0$

(B) Possible equation of C_1 is $x^2 + y^2 - 6x - 8y + 9 = 0$

(C) Radius of C_2 is 7

(D) Center of C_2 is $(-8, 1)$

Ans. ABC

Sol.

C02/2/017. Let $|X|$ denotes the cardinal number of set X and 'n' denote the number of possible values of p, such that $|A \cap B| = 3$, where $A = \{(x, y) \in \mathbb{R}^2 | x^2 + y^2 + 2x + 4y = p\}$, $B = \{(x, y) \in \mathbb{R}^2 | xy = 0\}$

Let $R(-1, 0)$, $S(1, 0)$ be ends of diameter of circle $C : x^2 + y^2 = 1$, and P be a variable point (other than R and S) be points on circle C , such that tangents to circle C at S and P meet at point Q . The normal to circle C at P intersects a line through Q parallel to RS at point E . Then the locus of E passes through:

(A) $\left(\frac{1}{2n-1}, \frac{1}{\sqrt{n+1}}\right)$

(B) $\left(\frac{1}{2n}, \frac{1}{n}\right)$

(C) $\left(\frac{1}{2n-1}, \frac{-1}{\sqrt{1+n}}\right)$

(D) $\left(\frac{1}{2n}, \frac{-1}{n}\right)$

Ans. AC

Sol.

C03/2/018. The tangent to the circle $x^2 + y^2 = 4$ at a point P intersect the parabola $y^2 = 4x$ at points Q and R . Tangents to the parabola at Q and R intersect at S . If Q lies in the first quadrant such that $PQ = 1$ unit, then :

(A) $PR = 8$ units

(B) $SR = \frac{35}{4}$ units

(C) $SQ = \frac{7}{2}$ units

(D) area of $\Delta SQR = \frac{343}{16}$ sq. units

Ans.

D

Sol.

$$OQ^2 = OP^2 + PQ^2$$

$$\Rightarrow t_1^2 + 4t_1^2 = 5$$

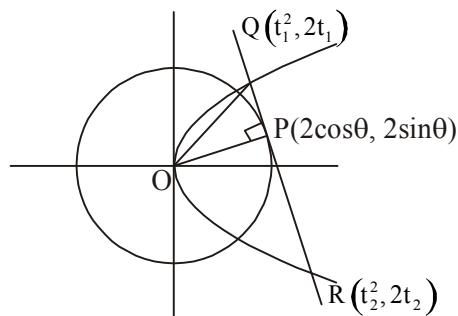
$$\Rightarrow t_1 = 1 (\because t_1 > 0)$$

$$\Rightarrow Q \equiv (1, 2)$$

$$PQ = 1 \quad \Rightarrow \cos\theta + 2\sin\theta = 2$$

$$\Rightarrow \cos\theta = \frac{4}{5}, \sin\theta = \frac{3}{5}$$

$$\Rightarrow P \equiv \left(\frac{8}{5}, \frac{6}{5}\right)$$



Now R lies on tangent $4x + 3y = 10$

$$\Rightarrow t_2 = \frac{-5}{2} (\because t_2 < 0)$$

$$\Rightarrow R \equiv \left(\frac{25}{4}, -5\right) \text{ and } S \equiv \left(-\frac{5}{2}, -\frac{3}{2}\right)$$

$$\text{Area of } \Delta SQR = \frac{343}{16} \text{ square unit}$$

C03/2/019. A parabola touches x-axis at origin such that foot of perpendicular from origin to axis of parabola is $N(-2, 2)$ then :

(A) eqn. of directrix is $x + y + 4 = 0$

(B) Focus of parabola is $(-2, 2)$

(C) Length of latus rectum is $2\sqrt{2}$

(D) eqn. of axis of parabola is $y = x + 4$

Ans.

ABD

Sol.

$$N(-2, 2)$$

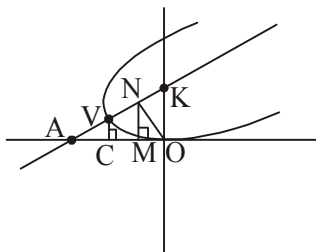
$$\angle NOM = 45^\circ$$

$$ON = AN = 2\sqrt{2}$$

$$OA = 4$$

$$NK = 2\sqrt{2} = 2a(\text{semi L.R.})$$

$$\Rightarrow a = \sqrt{2}$$



eqn. of NK : $y = x + 4$

$\therefore A : (-4, 0)$

$V : (-3, 1) \quad \{ \because (AV = VN) \}$

C03/2/020. Let ABCD be a rectangle with A(0, 0), B(4, 0), C(4, 4) and D(0, 4). Rectangle is folded in such a way that corner B always lies on line AD, then which of the following statement(s) is/are correct :

(A) As the point B moves on AD the crease thus formed will touch a fixed parabola whose focus is at (4, 0)

(B) As the point B moves on AD the crease thus formed will touch a fixed parabola whose focus is at

$$\left(\frac{3}{2}, 0 \right)$$

(C) As the point B moves on AD the crease thus formed will touch a fixed parabola whose equation is $y^2 = 8(x - 2)$

(D) As the point B moves on AD the crease thus formed will touch a fixed parabola whose equation of directrix is $y = 2x$.

Ans. AC

Sol. B(4, 0) is focus and $x = 0$ is directrix of parabola.

C03/2/021. Let $A_i(t_i^2, 2t_i)$; $t_i \neq 0$; $i = 1, 2, 3, 4$; be four points on parabola $y^2 = 4x$. Let (x_1, y_1) is the mid-point of focal chord A_1A_2 , and lines A_2A_3, A_1E are parallel, where $E(2, 0)$. If tangent at A_4 and normal at A_1 intersect at (x_2, y_2) , and $t_1t_4 = 1$, then :

(A) $t_3 = y_1$

(B) $t_3 = 2y_1$

(C) $x_2 = 1 + x_1 - t_2^2$

(D) $x_2 = 1 - x_1 + t_2^2$

Ans. AC

Sol.

C03/2/022. The common tangents to $C_1 : x^2 + y^2 = 2$ and $C_2 : y^2 = 8x$ touch the circle C_1 at P, Q and parabola C_2 at R, S. Also, normal of parabola $C_3 : y^2 = 4x$ drawn at ends of its latus-rectum are tangents to a circle C_4 with center $(3, -2)$ and radius r . then area of quadrilateral PQRS is equal to :

(A) $4r^2 + 7$

(B) $5r^2 + 5$

(C) $3r^2 + 7$

(D) $7r^2 - 2$

Ans. AB

Sol.

C04/2/023. Let $A_1(x_1, y_1)$ and $A_2(x_2, y_2)$ be foci of ellipse $E \equiv \frac{x^2}{9} + \frac{y^2}{8} = 1$; $x_1 < x_2$. Let P be a parabola having vertex at origin, focus at A_2 which intersects ellipse E at point M in first quadrant, and point N in fourth quadrant. Let tangents to ellipse E at M and N intersect at R, normal to parabola P at M intersects X-axis at Q. Which of the following options is/are correct if (x_3, y_3) are coordinates of ortho-center of ΔA_1MN and

$$\lambda = \frac{\text{Area of } \Delta MQR}{\text{Area of quadrilateral } MA_1NA_2} :$$

$$(A) 10x_3 + 8y_3 + 9 = 0$$

$$(B) 8\lambda = 5$$

$$(C) 9x_3 + 8y_3 + 10 = 0$$

$$(D) 5\lambda = 4$$

Ans. AB

Sol.

C04/2/024. Let A and A' be ends of major axis of ellipse $E \equiv x^2 + 2y^2 = 2$. Let P be any point on ellipse E. Let C be the locus of point of intersection of perpendiculars drawn from A and A' to A'P and AP respectively. Let (x_1, y_1) , (x_2, y_2) , be foci of C, (x_3, y_3) be a point of intersection of C and E, $ax + y + b = 0$ and ℓ be the equation of directrix and length of latus-rectum of C. Then which of the following options is/are correct :

$$(A) x_1^2 + 2x_2^2 + 3x_3^2 = 4$$

$$(B) \ell = 2$$

$$(C) 3x_1^2 + 2x_2^2 + x_3^2 = 2$$

$$(D) b^2 = 8$$

Ans. BCD

Sol.

C04/2/025. Let E be an ellipse with major and minor axis of length 4 and $2\sqrt{3}$ respectively. Let P be any point on E, and S, S' are foci of E. Let C be the center of E, foot of perpendicular from P on major axis is P', normal at P cuts major axis at N and λ denote the length of portion of tangent at P intercepted between principal axes. Then which of the following options is/are correct :

$$(A) PS \cdot PS' \leq 4$$

$$(B) PS^2 + PS'^2 \geq 8$$

$$(C) \lambda \geq 2 + \sqrt{3}$$

$$(D) 4CN = CP'$$

Ans. ABCD

Sol.

C04/2/026. Let $m_1 < m_2 < m_3$ be the slopes of distinct normal of ellipse E passing through point A, where

$$E \equiv \frac{x^2}{16} + \frac{y^2}{12} = 1, A = \left(\frac{1}{2\sqrt{2}}, \frac{\sqrt{2}}{2\sqrt{3}} \right) \text{ Then which of the following options is/are correct :}$$

$$(A) m_1 < m_2 < m_3 = 2\sqrt{3}$$

$$(B) 3\sqrt{3}m_1m_2m_3 = -8$$

$$(C) \sqrt{3}(m_1 + m_2 - m_3) = 4\sqrt{3} + 2$$

$$(D) \sqrt{3}(m_3 - m_2 - m_1) = 4\sqrt{3} - 2$$

Ans. AB

Sol.

C04/2/027. An ellipse has focus at origin, and passes through P (4, 3) and Q (−4, −3). Length of major axis is 2a, and

that of minor axis is $2b$. (x_1, y_1) is its centre and (x_2, y_2) is the other focus. If eccentricity of ellipse is $\frac{1}{\sqrt{2}}$ and $y_1 > 0$, then :

(A) $x_1 + y_1 = \sqrt{2}$

(B) $x_2 + y_2 = 3\sqrt{2}$

(C) $b = 5\sqrt{2}$

(D) $a = 20$

Ans. AC

Sol.

C04/2/028. Which of the following options is / are correct for ellipse $E \equiv \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$:

(A) Foot of perpendicular from focus of E on any tangent lies on its auxiliary circle.

(B) Perpendicular from focus of E on any tangent, and line from center to corresponding point of contact of tangent intersect on corresponding directrix.

(C) If tangent at any point of ellipse E makes angle α with major and β with focal radius, then eccentricity of E is $e = \left| \frac{\cos \alpha \cos \beta}{\cos \alpha} \right|$.

(D) If E is the ellipse having least possible area, and also containing the circle $(x - 1)^2 + y^2 = 1$, then eccentricity of E is $e = \sqrt{\frac{2}{3}}$.

Ans. ABCD

Sol.

V01/2/029. The vertices of a triangle ABC are $A = (2, 0, 2)$, $B = (-1, 1, 1)$ and $C = (1, -2, 4)$. The points D and E divide the sides AB and CA in the ratio 1 : 2 respectively. Another point F is taken in space such that the perpendicular drawn from F to the plane containing $\triangle ABC$, meets the plane at the point of intersection of the line segments CD and BE. If the distance of F from the plane of triangle ABC is $\sqrt{2}$ units, then :

(A) The volume of the tetrahedron ABCF is $\frac{7}{3}$ cubic units

(B) The volume of the tetrahedron ABCF is $\frac{7}{6}$ cubic units

(C) One of the equation of the line AF is $\vec{r} \cdot (2\hat{i} + 2\hat{k}) + \lambda(2\hat{k} - \hat{i}) (\lambda \in \mathbb{R})$

(D) One of the equation of the line AF is $\vec{r} \cdot (2\hat{i} + 2\hat{k}) + \mu(\hat{i} + 7\hat{k})$

Ans. AC

Sol.

$$CD : \vec{r} = (\hat{i} - 2\hat{j} + 4\hat{k}) + \frac{\lambda}{3}(7\hat{j} - 7\hat{k})$$

$$BE : \vec{r} = (-\hat{i} + \hat{j} + \hat{k}) + \frac{\mu}{3}(7\hat{i} - 7\hat{j} + 7\hat{k})$$

$$P = (\hat{i} - \hat{j} + 3\hat{k})$$

Volume of tetrahedron ABCF

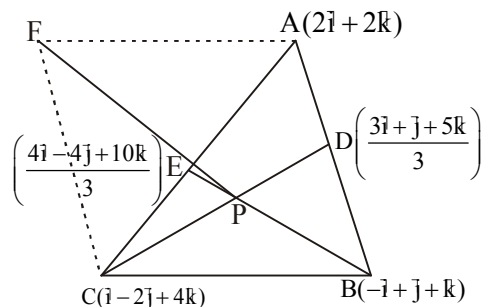
$$= \frac{1}{3} (\text{Area of base triangle}) \times \text{height} = \frac{7}{3} \text{ cubic units}$$

$$\overrightarrow{AB} \times \overrightarrow{AC} = 7\hat{i} + 7\hat{k}, |\overrightarrow{PF}| = PF = \sqrt{2} \text{ units}$$

$$\overrightarrow{PF} = \sqrt{2} \left(\frac{7\hat{j} + 7\hat{k}}{\sqrt{49 + 49}} \right) = \hat{j} + \hat{k} = \text{P.V. of F} - \text{P.V. of P}$$

$$\text{P.V. of F} = F = \hat{i} + 4\hat{k}$$

$$\text{Vector equation of AF is } \vec{r} = 2(\hat{i} + \hat{k}) + a(-\hat{i} + 2\hat{k})$$



V01/2/030. If $\vec{p}, \vec{q}, \vec{r}$ are three non-coplanar vectors satisfying $x(-\vec{p} + 2\vec{q} + 3\vec{r}) + y(2\vec{p} + 3\vec{q} - 4\vec{r}) + z(\vec{p} - 4\vec{q} + 5\vec{r}) = -4\vec{p} - 2\vec{q} + 38\vec{r}$, then choose the correct option (s) :

(A) $5x - y + z = 17$

(B) $2x - 2y + 6z = 33$

(C) $x + 5y - 3z = 25$

(D) $4x + y + 2z = 13$

Ans.

AD

Sol.

V01/2/031. Let $\vec{a}, \vec{b}, \vec{c}$ be vectors in $\mathbb{R}^3$ satisfying $|\vec{a}| = 1 = |\vec{b}|$, $\frac{\vec{b} \cdot \hat{i}}{1} = \frac{\vec{b} \cdot \hat{j}}{1} = \frac{\vec{b} \cdot \hat{k}}{2}$, $|\vec{a} \times \vec{c}| = 1$, $\vec{a} \cdot (\vec{b} \times \vec{c}) = 1$. Then which of the following options is /are correct :

(A) There exists exactly one choice for $\vec{c}$.

(B) There exists infinitely many choices for $\vec{c}$

(C) If $\vec{a}$ lies on xy-plane, then $|\vec{a} \cdot \hat{i}| = |\vec{a} \cdot \hat{j}|$

(D) If $\vec{a}$ lies xz-plane, then $2|\vec{a} \cdot \hat{i}| = |\vec{a} \cdot \hat{k}|$.

Ans.

BC

Sol.

V01/2/032. Let $\vec{a}, \vec{b}, \vec{c}$ are three-unit vectors satisfying $|\vec{2a} - \vec{b}|^2 + |\vec{2b} - 3\vec{c}|^2 + |\vec{c} - 3\vec{a}|^2 = 42$ Then which of the following options is / are correct :

(A) $|\vec{12a} + 2\vec{b} + 3\vec{c}| = 11$

(B) $|\vec{12a} + 4\vec{b} + 6\vec{c}| = 10$

(C) $|\vec{12a} + 6\vec{b} + 9\vec{c}| = 9$

(D) $|\vec{12a} + 16\vec{b} + 24\vec{c}| = 5$

Ans. ABC

Sol.

V01/2/033. Which of the following options is/are correct :

(A) Let $\vec{p}_r = a_r \hat{i} + b_r \hat{j} + c_r \hat{k}, |\vec{p}_r| = 1; r = 1, 2, 3$, then there exist infinite elements in the set

$$\{(a_1, b_1, c_1, a_2, b_2, c_2, a_3, b_3, c_3) : \vec{p}_1 = \vec{p}_2 \times \vec{p}_3 - \vec{p}_1 \times \vec{p}_2\}.$$

(B) Distance of ortho- center of triangle formed by (3,4,2), (3,0,2) and (5,0,0) from origin is $\sqrt{13}$

(C) If $a, b, c \in \mathbb{R}$, satisfying $a^2 + b^2 + c^2 = 1$, then maximum value of $\max(13a^2 + 10b^2 + 5c^2 - 4ab - 12bc - 6ac) = 14$.

(D) Let dir's of lines L, M and N are (1, 0, 0), (1, -1, 1), (a, b, c) respectively. Line M is bisector of angle between lines. L and N, if $3b + c = 4a$.

Ans. BCD

Sol.

V01/2/034. Given a line L and a plane $\Pi : L \equiv \frac{x+2}{2} = \frac{y+1}{-1} = \frac{z}{3}, \Pi \equiv x + y + z = 3$: Let lines L_1 and L_2 respectively be projection and image of line L in plane Π . Let planes Π_1, Π_2 respectively be planes containing L and parallel, perpendicular to Π . Then shortest distance of :

(A) Π_1 from origin is $\sqrt{3}$

(B) L_1 from $(-9, 6, -3)$ is $\sqrt{53}$

(C) L_2 from origin is $\sqrt{\frac{285}{7}}$

(D) Π_2 from origin is $\sqrt{\frac{49}{26}}$

Ans. ABD

Sol.

COMPREHENSION TYPE QUESTIONS

Paragraph for Question C02/3/035 to C02/3/036

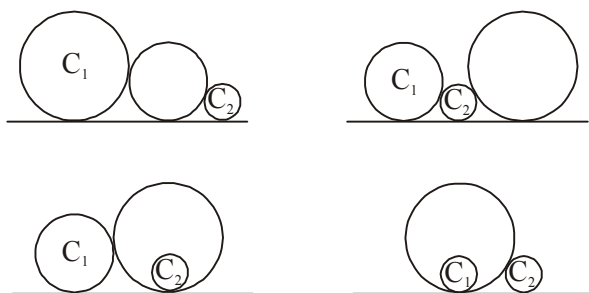
The centres of two circles C_1 and C_2 of radii 4 and 9 respectively are at a distance of 15 units from each other. Let L_1 be one direct common tangent to these circles.

C02/3/035. There are n circles which touches both the circle C_1 and C_2 and also the line L_1 , then n is :

- (A) 2 (B) 4 (C) 3 (D) 8

Ans. B

Sol.

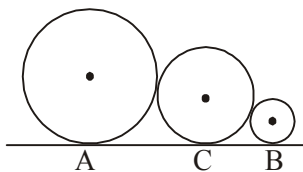


C02/3/036. Let $r_1, r_2, \dots, r_n$ be the radii of these circles then find $\max(r_1, r_2, \dots, r_n)$:

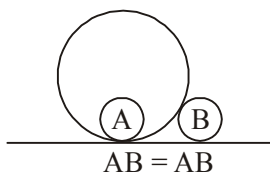
- (A) $\frac{25}{2}$ (B) $\frac{50}{9}$ (C) 50 (D) 1

Ans. C

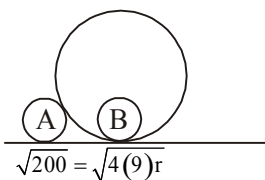
Sol. $AB = AC + BC$



$$= \sqrt{225 - 25} = \sqrt{4r(9)} + \sqrt{4r(4)}$$



$$AB = AB$$

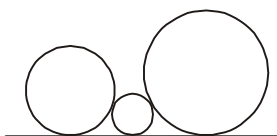


$$\sqrt{200} = \sqrt{4(9)r}$$

$$\sqrt{4r(4)} = \sqrt{200}, r = \frac{50}{9}$$

$$r = \frac{25}{2}$$

$$AC - BC = AB$$



$$\sqrt{4(9)r} - \sqrt{4(4)r} = \sqrt{200}$$

$$2\sqrt{r} = \sqrt{200}$$

$$r = 50$$

Paragraph for Question C02/3/037 to C02/3/038

Let ABC be a triangle such that origin is its circumcentre and its base, BC, always passes through the point (4, 5). If the lateral sides (AB and AC) have $x + y = 0$ and $x - 9y = 0$ as their perpendicular bisectors, then the locus of A is a circle L, then,

C02/3/037. L passes through which of the below points :

- (A) (0, 0) (B) (1, 41) (C) $\left(\frac{4}{5}, \frac{3}{5}\right)$ (D) (-1, -2)

Ans. A

Sol.

C02/3/038. The centre of L lies on the line :

- (A) $x + 5y = 7$ (B) $x - y = 0$ (C) $x = 0$ (D) $y = 0$

Ans. B

Sol.

Let A (h, k), C (x_0, y_0), B (x_1, y_1)

$$h + x_0 = 9(k + y_0) \text{ and } h^2 + k^2 = x_0^2 + y_0^2$$

$$\Rightarrow x_0 = \frac{1}{41}(40h + 9k), y_0 = \frac{1}{41}(9h - 40k)$$

$$h + x_1 = -(k + y_1) \text{ and } h^2 + k^2 = x_1^2 + y_1^2 \quad \Rightarrow x_1 = -k, y_1 = -h$$

$$\frac{5 - y_1}{4 - x_1} = \frac{5 - y_0}{4 - x_0} \quad \Rightarrow 4h^2 + 4k^2 + 41k = 0$$

Paragraph for Question C02/3/039 to C02/3/040

Let points $A_i; i \equiv 1, 2, 3, 4, 5, 6$ lying on circle $S \equiv x^2 + y^2 = 4$, satisfy the following conditions. Chords $A_1 A_2$ and $A_3 A_4$, of S pass through $P_0(1, 1)$ and are parallel to X-axis and Y-axis respectively.

$A_5 A_6$ is chord of S passing through P_0 and having slope -1.

Tangents to S at A_1, A_2 intersect at A_7 , the tangents of S at A_3, A_4 meet at A_8 and the tangents of S at A_5, A_6 meet at A_9 . P is a point on circle S in first quadrant and tangent of S at P intersects coordinate axes at M and N.

Now, based on above information, answer the following questions

C02/3/039. The points A_7, A_8, A_9 lie on curve :

- (A) $(x - 4)^2 + (y - 4)^2 = 16$ (B) $x + y = 4$
(C) $(x - 4)(y - 4) = 4$ (D) $xy = 4$

Ans. B
Sol.

C02/3/040. Mid-point of MN lies on the curve :

- (A) $(x + y)^2 = 3xy$ (B) $x^{2/3} + y^{2/3} = 2^{4/3}$
(C) $x^2 + y^2 = 2xy$ (D) $x^2 + y^2 = x^2 y^2$

Ans. D
Sol.

Paragraph for Question C03/3/041 to C03/3/042

Let $a, r, s, t \in \mathbb{R} - \{0\}$, and $P(at^2, 2at)$, $Q, R(ar^2, 2ar)$, $S(as^2, 2as)$ are distinct points on parabola $y^2 = 4ax$. Suppose that PQ is focal chord, and lines QR, PK are parallel where $K(2a, 0)$. Let tangent at P and normal at S to parabola meet at $T(\alpha, \beta)$. If curve C is the mirror image of parabola w.r.t. line L_1 , and A, B are the points of intersection of C with line L_2 , where $L_1: x + y + 4 = 0$ and $L_2: y + 5 = 0$. Now, based on above information, answer the following questions.

C03/3/041. If $t = 4$, then $(r^2) \cdot AB$ is equal to :

- (A) 13 (B) 14 (C) 15 (D) 16

Ans. C
Sol.

C03/3/042. If $a = 5t = 125s = 25$, then $\sqrt{2.5\beta}$ is equal to :

- (A) 10 (B) 11 (C) 13 (D) 14

Ans. C
Sol.

Paragraph for Question C03/3/043 to C03/3/044

Let us consider the following parabolas: $\Pi_1: y^2 = 2x$ and $\Pi_2: y^2 = 4x$. Let A and B are distinct points on parabola Π_1 , such that circle with AB as diameter passes through its vertex O, such that A lies in first quadrant and area of ΔOAB is $3\sqrt{2}$ square units.

Let S be focus of parabola Π_2 , and pair of tangents drawn from $P(-2, 1)$ to Π_2 , touch it at P_1 and P_2 . Let Q_1, Q_2 be points on lines SP_1, SP_2 respectively such that PQ_1 is perpendicular to SP_1 and PQ_2 is perpendicular to SP_2 . Now, based on above information, answer the following questions

C03/3/043. Which of the following is/are coordinates of A :

- (A) $(4, 2\sqrt{2})$ (B) $(9, 3\sqrt{2})$ (C) $\left(\frac{1}{4}, \frac{1}{\sqrt{2}}\right)$ (D) $(1, \sqrt{2})$

Sol.

C03/3/044. Which of the following options is/are correct :

- (A) $SQ_1 = 2$ (B) $5Q_1Q_2 = 3\sqrt{10}$ (C) $PQ_1 = 3$ (D) $SQ_2 = 1$

Ans. BCD

Sol.

Paragraph for Question C01/3/045 to C01/3/046

Diameter of ellipse is a line passing through the centre of the ellipse. Consider the set of parabola(s) having common chord of minimum length with the diameter of the ellipse $x^2 + y^2 + xy = 8$ and the line $mx - y + 4 = 0$ as directrix.

C04/3/045. The set of value(s) of m for which only one parabola is possible is :

- (A) $\{-\sqrt{2}, \sqrt{2}\}$ (B) $\{-\sqrt{5}, \sqrt{3}\}$ (C) $\{-2, 2\}$ (D) None of these

Ans. A

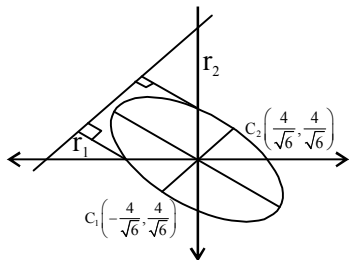
Sol.

C04/3/046. The range of value (s) of m for which two parabola(s) are possible is :

- (A) $(-\infty, -\sqrt{2}) \cup (\sqrt{2}, \infty)$ (B) $(-\infty, -2) \cup (2, \infty)$
(C) $(-\infty, -\sqrt{3}) \cup (\sqrt{3}, \infty)$ (D) None of these

Ans. A

Sol.



For the ellipse $x^2 + y^2 + xy = 8$, $C \equiv (0, 0)$

For the parabola $\frac{8}{\sqrt{m^2 + 1}} = \frac{8}{\sqrt{3}}$: ($\because C_1 C_2 = r_1 + r_2$)

For two parabola (s) : $C_1 C_2 < r_1 + r_2$

Paragraph for Question C05/3/047 to C05/3/048

Let circle C and hyperbola H intersect at points A and B. Let S be focus, L be the corresponding directrix

of ellipse E. P is a parabola with focus at S and line L as directrix, where $C : x^2 + y^2 = 8x$, $E : \frac{x^2}{4} + \frac{y^2}{3} = 1$,

$H : \frac{x^2}{9} - \frac{y^2}{4} = 1$. Now, based on above information, answer the following questions.

C05/3/047. If number of common tangents between P and E is n, and $ax + by + 4 = 0$ is equation of common tangents with positive slope to C and H, then $n + a + b^2$ is equal to :

- (A) 9 (B) 8 (C) 7 (D) 6

Ans. A

Sol.

C05/3/048. If m denotes the number of normals between P and E, and $x^2 + y^2 + 2gx + 2fy + c = 0$ is the equation of circle with AB as diameter, then $2g + 2f + 2c + m$ is equal to :

- (A) 36 (B) 38 (C) 40 (D) 42

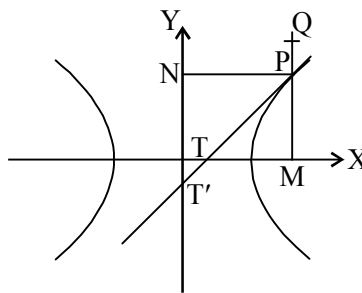
Ans. B

Sol.

Paragraph for Question C05/3/049 to C05/3/050

Figure shows a hyperbola whose centre is C. Tangent is drawn at a variable point 'P' in Ist quadrant meeting transverse and conjugate axes at T and T' respectively. M and N are feet of perpendiculars from P on transverse and conjugate axes as shown. Q is a point on ordinate MP produced such that $MQ = SP$ where S is the focus. $CM \cdot CT = 4$ and $CN \cdot CT' = 5$.

On the basis of above information, answer the following question :



C05/3/049. Eccentricity of hyperbola is :

- (A) $\sqrt{2}$ (B) $\frac{3}{2}$ (C) 3 (D) 2

Ans. B

Sol.

C05/3/050. If locus of Q is straight line, then its :

(A) x-intercept is $\frac{3}{4}$

(B) x-intercept is -2

(C) Slope is $\frac{3}{2}$

(D) Slope is $-\frac{3}{2}$

Ans. BC

Sol.

INTEGER TYPE QUESTIONS

C01/4/051. Let three lines L_1, L_2 and L_3 belonging to the family $x - 2y + 6 + \lambda(x - y + 2) = 0$, where λ is a parameter, be interior angle bisectors of ΔABC . If the equation $x + 3y - 4 = 0$ represents side AB of the triangle, then

the value of $\left[\frac{\Delta}{\sum \left(r \cot \frac{A}{2} + a \right)} \right]$ is, _____.

(Note : Symbols used have usual meaning in ΔABC and $[.]$ denotes G.I.F.)

Ans. 1

Sol. Intersection point of $x - 2y + 6 = 0$ and $x - y + 2 = 0$ is the incentre of the ΔABC $I(2,4)$

$r =$ in radius = Perpendicular distance of I from the line AB

$$= \left| \frac{2 + 12 - 4}{\sqrt{10}} \right|, r = \sqrt{10}$$

$$\sum \left(r \cot \frac{A}{2} + a \right) = r \cot \frac{A}{2} + a + r \cot \frac{B}{2} + b + r \cot \frac{C}{2} + c$$

$$= s - a + a + s - b + b + s - c + c$$

$$\left[\frac{\Delta}{\sum \left(r \cot \frac{A}{2} + a \right)} \right] = \left[\frac{rs}{3s} \right] = \left[\frac{\sqrt{10}}{3} \right] = 1$$

C01/4/052. ΔABC is an equilateral triangle with circumcentre $(1, 2)$ and vertex A lying on the line $5x + 2y = 4$. A circle with centre $I(2, 0)$ passes through the vertices B and C and intersects the sides AC and AB at D and E

respectively. If area of quadrilateral BCDE is Δ , then find the value of $\frac{\Delta}{10\sqrt{3}}$ is _____.

Ans. 9

Equation of OI is, $2x + y = 4$

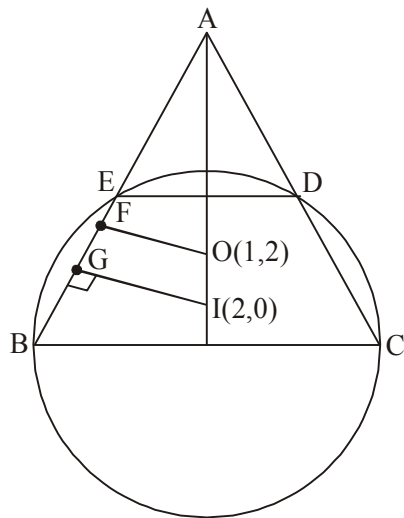
Vertex is point intersection of $2x + y = 4$ and $5x + 2y = 4$

$$\therefore A = (-4, 12), AO = 5\sqrt{5}, AI = 6\sqrt{5}$$

$$\therefore AG = \frac{AE + AB}{2}$$

$$\therefore AE = 2AG - AB = \sqrt{15}$$

$$\therefore \text{Area of quadrilateral BCDE} = \text{area of } \triangle ABC - \text{area of } \triangle ADE$$



$$= \frac{\sqrt{3}}{4} (AB^2 - AE^2) = 90\sqrt{3}$$

C01/4/053. Let ABC be right angle triangle, $\angle B = 90^\circ$. The median through A and C are $y = 3x + 1$ and $y = x + 1$ respectively. If $AC = 8$, then area of triangle ABC is _____.

Ans. 10.66

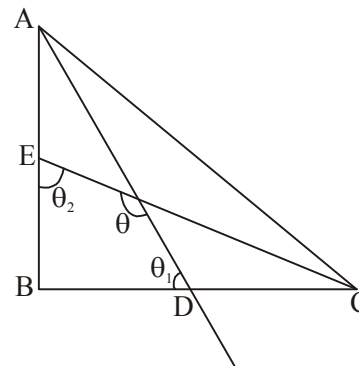
Sol. $\tan \theta_1 = \frac{AB}{BD} = \frac{2c}{a}$

$$\tan \theta_2 = \frac{BC}{BE} = \frac{2a}{c}$$

$$\tan\theta_1 \cdot \tan\theta_2 = 4$$

$$\text{Angle between median} = \tan^{-1}\left(\frac{3-1}{1+3}\right) = \tan^{-1} \frac{1}{2}$$

$$\theta = \pi - \tan^{-1} \frac{1}{2}$$



$$\text{Now, } \theta_1 + \theta_2 + \theta = \frac{3\pi}{2}$$

$$\theta_1 + \theta_2 + \pi - \tan^{-1} \frac{1}{2} = \frac{3\pi}{2}$$

$$\theta_1 + \theta_2 = \frac{\pi}{2} + \tan^{-1} \frac{1}{2}$$

$$\tan(\theta_1 + \theta_2) = \tan\left(\frac{\pi}{2} + \tan^{-1} \frac{1}{2}\right) = -\cot\left(\tan^{-1} \frac{1}{2}\right) = -2$$

$$\frac{\tan \theta_1 + \tan \theta_2}{1 - \tan \theta_1 \tan \theta_2} = -2$$

$$\tan \theta_1 + \tan \theta_2 = -2(1 - 4) = 6$$

$$\frac{2c}{a} + \frac{2a}{c} = 6 \Rightarrow \frac{c^2 + a^2}{ac} = 3 \Rightarrow \frac{64}{ac} = 3 \Rightarrow ac = \frac{64}{3}$$

$$\text{Area} = \frac{1}{2}ac = \frac{32}{3}$$

C01/4/054. Coordinate of any point P is (4, 5). If A and B are variable points on straight lines $y = x$ and $y = 2x$ respectively. Then find minimum value of $(PA + PB + AB)^2$ is _____.

Ans. 16.40

Sol. Reflection of (4, 5) w.r.t. lines $y = x$ and $y = 2x$ are (5, 4) and $\left(\frac{8}{5}, \frac{31}{5}\right)$ respectively. Minimum value of

$$PA + PB + AB = \text{distance between } (5, 4) \text{ and } \left(\frac{8}{5}, \frac{31}{5}\right) = \sqrt{16.4}$$

C01/4/055. If $|x| = \max \{a \in \mathbb{Z} : a \leq x\}$, then find the minimum value of $\left| \sqrt{x^2 - 2x + 2} + \sqrt{x^2 - 4x + 29} \right|$ _____.

Ans. 6

Sol.

C01/4/056. Find the value of $4x$, if λ is minimum, where

$$\lambda = AB + BC + CA ; A(5, 3), B(\alpha, \alpha), C(x, 0) \text{ _____.}$$

Ans. 17

Sol.

C01/4/057. Find the area of pentagon formed by points lying in $A \cup B$, where

$$A = \left\{ (x, y) \in \mathbb{Z}^2 \mid \det \begin{pmatrix} x^2 & 4-y \\ y-4 & 1 \end{pmatrix} \begin{pmatrix} x-3 & y \\ y & 3-x \end{pmatrix} \begin{pmatrix} x^2+y^2+37 & 2 \\ 6x+y & 1 \end{pmatrix} = 0 \right\},$$

$$B = \left\{ (x, y) \in \mathbb{Z}^2 \mid \det \begin{pmatrix} |x-4| & -1 \\ |y-9| & 1 \end{pmatrix} \begin{pmatrix} x-6 & y-1 \\ y-1 & 6-x \end{pmatrix} \begin{pmatrix} y-5 & 7-x \\ 7-x & 5-y \end{pmatrix} = 0 \right\}$$

Here $\det(A)$ denotes the determinant of matrix A _____.

Ans. 36.5

Sol.

C01/4/058. If value of the expression

$$\left(\sqrt{(x-2)^2 + (y-4)^2} - \sqrt{(x+2)^2 + (y-2)^2} \right)^4 + \left(\sqrt{(x-2)^2 + (y-4)^2} + \sqrt{(x+2)^2 + (y-2)^2} \right)^4$$

is minimum for $x = x_0$ and $y = y_0$, then $|x_0 + y_0|$ is equal to _____.

Ans. 3

Sol.

C02/4/059. Suppose that the circle $x^2 + y^2 - 4ax - 2ay + 4a^2 = 0$ ($a > 0$) touches the lines $y = mx$ at A and B such that

$AB = 3$. Then the value of $\frac{4a}{\sqrt{5}}$ is _____.

Ans. 3

Sol. $y = mx$ touches circle

$$\Rightarrow D = 0 \text{ for } (m^2 + 1)x^2 - 2a(m+1)x + 4a^2 = 0$$

$$\Rightarrow 4a^2(m+1)^2 = 4(m^2+1)(4a^2)$$

$$\Rightarrow m = 0, \frac{4}{3}$$

Let $A : (x_1, m_1x_1)$, $B : (x_2, m_2x_2)$

$$AB^2 = 9 \Rightarrow (x_1 - x_2)^2 + (m_1x_1 - m_2x_2)^2 = 9 \text{ where}$$

$$x_1 = \frac{(m_1+2)}{m_1^2+1}a, x_2 = \frac{(m_2+2)}{m_2^2+1}a, m_1 = 0, m_2 = \frac{4}{3}$$

$$\text{get } a = \frac{3\sqrt{5}}{4}$$

C02/4/060. Two circles C_1, C_2 of radii $\frac{1}{3}$ and $\frac{2}{7}$ touch each other externally and they both touch a unit circle C internally. A circle C_3 of radius R is inscribed to touch the circles C_1, C_2 externally and the circle C internally, then $19R$ is equal to _____.

Ans. 2

Sol.
$$R = \frac{\frac{1}{3} \cdot \frac{2}{7}}{1 - \frac{1}{3} \cdot \frac{2}{7}} = \frac{2}{19}$$

C02/4/061. A ball moving around the circle $x^2 + y^2 - 2x - 4y - 20 = 0$ in anticlockwise direction leaves it tangentially at $P(-2, -2)$. After getting reflected from a straight line 'L', it passes through the centre of circle. If distance of P from line L is $\frac{5}{2}$ and 2α be the angle between incident ray and reflected ray, then the value of $10 \cot 2\alpha \cos \alpha$ is equal to _____.

Ans. 5

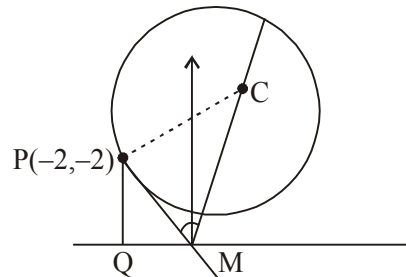
Sol.
$$\tan 2\alpha = \frac{PC}{PM} = \frac{5}{PM} \quad \text{_____ (1)}$$

$$\sin(90 - \alpha) = \frac{PQ}{PM}$$

$$PM = \frac{5/2}{\cos \alpha} = \frac{5}{2 \cos \alpha} \quad \text{_____ (2)}$$

$$5 \cot 2\alpha = \frac{5}{2 \cos \alpha}$$

$$\Rightarrow 10 \cot 2\alpha \cos \alpha = 5$$



C02/4/062. Given circles C_1 and C_2 intersect at X and Y . Let I_1 be a line through the centre of C_1 and intersecting C_2 at points P & Q and let I_2 be the line through centre of C_2 intersecting C_1 at pts R and S . Let $C_1 : x^2 + y^2 - 2x - 2y + 1 = 0$ and $C_2 : x^2 + y^2 - 4x - 4y + 4 = 0$. If P, Q, R, S are concyclic with centre of circle passing through them being (α, β) , then $\tan^{-1}(\alpha + \beta) + \tan^{-1}(\alpha + \beta - 1)$ is equal to $\tan^{-1}\left(\frac{10}{K}\right)$, then K is _____.

Ans. 1.25

Sol. Conceptual

C04/4/063. If ℓ_1 and ℓ_2 be the length of the focal chord other than latus rectum of ellipse $\frac{x^2}{4} + \frac{y^2}{3} = 1$ which are perpendicular to each other, then $12\left(\frac{1}{\ell_1} + \frac{1}{\ell_2}\right)$ _____.

Ans. 7

Sol. Let AB = I,

$$\text{Equation of AB} = \frac{x-1}{\cos \theta} = \frac{y-0}{\sin \theta} = r$$

$$\Rightarrow \frac{(1+r \cos \theta)^2}{4} + \frac{(r \sin \theta)^2}{3} = 1$$

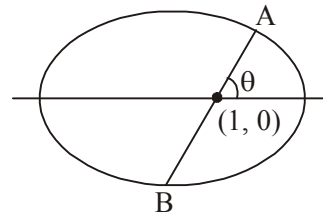
$$\Rightarrow r^2 (3 + \sin^2 \theta) + 6r \cos \theta - 9 = 0$$

$$\Rightarrow r_1 - r_2 = -\frac{6 \cos \theta}{3 + \sin^2 \theta}, r_1 r_2 = \frac{9}{3 + \sin^2 \theta}$$

$$I_1 = |r_1| + |r_2| = \sqrt{(r_1 - r_2)^2 + 4r_1 r_2} = \frac{12}{3 + \sin^2 \theta}$$

$$I_2 = \frac{12}{3 + \sin^2 \left(\frac{\pi}{2} + \theta\right)} = \frac{12}{3 + \cos^2 \theta}$$

$$= \frac{1}{I_1} + \frac{1}{I_2} = \frac{7}{12}$$



C04/4/064. Two tangents to an ellipse $x^2 + 2y^2 = 2$ are drawn which intersect at right angles, let ℓ_1 and ℓ_2 be the intercepts on these tangents made by the circle $x^2 + y^2 = 2$, then $\ell_1^2 + \ell_2^2$ is equal to _____.

Ans. 4

Sol. As $\ell_1^2 + \ell_2^2 = (SS_1)^2$ (S, S_1 are foci of ellipse)

C04/4/065. Consider the ellipse $4x^2 + y^2 - 8x + 4y = 0$. Pair of tangents PT_1 and PT_2 are drawn to the ellipse from $P(-8, -2)$. Let S'_1 be the image of S_1 about PT_1 and S'_2 be the image of S_2 about PT_2 , then if

$\angle PS_1 S'_2 = \alpha$ and $\angle PS'_1 S_2 = \beta$, then $\frac{\alpha}{\beta}$ is equal to _____.

Ans. 1

Sol. $\Delta PS'_1 S_2$ and $\Delta PS_1 S'_2$ are congruent.

C04/4/066. The straight line $\frac{x}{4} + \frac{y}{3} = 1$ intersects the ellipse $\frac{x^2}{16} + \frac{y^2}{9} = 1$ at two points A and B, there is a point P on this ellipse such that the area of ΔPAB is equal to $6(\sqrt{2} - 1)$, then the number of such points (P) is _____.

Ans. 3

Sol. Let perpendicular distance of P from the line be h.

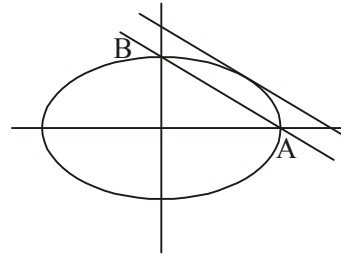
$$\frac{1}{2} \times h \times 5 = 6(\sqrt{2} - 1) \text{ (as } \Delta PAB = 6(\sqrt{2} - 1))$$

$$\Rightarrow h = \frac{12(\sqrt{2} - 1)}{5}$$

Now distance of tangent parallel to AB i.e.,

$$4y + 3x = 12\sqrt{2}, \text{ from line AB is } \frac{12(\sqrt{2} - 1)}{5}$$

There are just three such points.



C04/4/067. Let M_{AB} denote the mid-point of line-segment AB. Let P, Q, R be distinct points on ellipse $E \equiv \frac{x^2}{16} + \frac{y^2}{9} = 1$ and $d = (\text{distance between } M_{PQ} \text{ and } M_{PR})$. Then maximum possible value of d is equal to _____.

Ans. 4

Sol.

C05/4/068. If Δ is the area of parallelogram formed by common tangents of ellipse E and hyperbola H, and λ is the smallest distance of any point on ellipse E' from its center, then find $(b - a)\Delta$, where

$$E \equiv \frac{x^2}{4} + y^2 = 1, H \equiv \frac{x^2}{5} - \frac{y^2}{3} = 1, E' \equiv 5x^2 + 4y^2 + xy = 2, \lambda = \frac{a}{\sqrt{b - \sqrt{2}}}. \text{ _____}$$

Ans. 119

Sol.

V01/4/069. If $\vec{a}_1 = a\hat{i} + b\hat{j}$, $\vec{a}_2 = \vec{a}_1 - (2\hat{i} + \sqrt{5}\hat{j})$ and $\vec{a}_3 = \vec{a}_1 + (2\hat{i} - \sqrt{5}\hat{j})$, then $\begin{vmatrix} 1 + \vec{a}_1 \cdot \vec{a}_1 & 1 + \vec{a}_1 \cdot \vec{a}_2 & 1 + \vec{a}_1 \cdot \vec{a}_3 \\ 1 + \vec{a}_2 \cdot \vec{a}_1 & 1 + \vec{a}_2 \cdot \vec{a}_2 & 1 + \vec{a}_2 \cdot \vec{a}_3 \\ 1 + \vec{a}_3 \cdot \vec{a}_1 & 1 + \vec{a}_3 \cdot \vec{a}_2 & 1 + \vec{a}_3 \cdot \vec{a}_3 \end{vmatrix}$ is equal to _____.

Ans. 80

Sol. $\vec{a}_1 = (a, b), \vec{a}_2 = (a - 2, b - \sqrt{5}), \vec{a}_3 = (a + 2, b - \sqrt{5})$

Area of Δ , whose vertices are $\vec{a}_1, \vec{a}_2$ and $\vec{a}_3$, is $\Delta = \frac{1}{2} \begin{vmatrix} 1 & a & b \\ 1 & a-2 & b-\sqrt{5} \\ 1 & a+2 & b-\sqrt{5} \end{vmatrix}$

$$\Rightarrow \Delta^2 = \frac{1}{4} \begin{vmatrix} 1 & a & b \\ 1 & a-2 & b-\sqrt{5} \\ 1 & a+2 & b-\sqrt{5} \end{vmatrix} \begin{vmatrix} 1 & a-2 & b-\sqrt{5} \\ 1 & a+2 & b-\sqrt{5} \end{vmatrix} \text{ and } |\vec{a}_1 - \vec{a}_2| = 3, |\vec{a}_1 - \vec{a}_3| = 3, |\vec{a}_2 - \vec{a}_3| = 4$$

$$\Rightarrow \Delta = \sqrt{5 \times 2 \times 2 \times 1} = 2\sqrt{5}$$

$$(2\sqrt{5})^2 = \frac{1}{4} \begin{vmatrix} 1 + \vec{a}_1 \cdot \vec{a}_1 & 1 + \vec{a}_1 \cdot \vec{a}_2 & 1 + \vec{a}_1 \cdot \vec{a}_3 \\ 1 + \vec{a}_2 \cdot \vec{a}_1 & 1 + \vec{a}_2 \cdot \vec{a}_2 & 1 + \vec{a}_2 \cdot \vec{a}_3 \\ 1 + \vec{a}_3 \cdot \vec{a}_1 & 1 + \vec{a}_3 \cdot \vec{a}_2 & 1 + \vec{a}_3 \cdot \vec{a}_3 \end{vmatrix}$$

$$\therefore \text{Required answer} = 4 \times 5 \times 4 = 80$$

V01/4/070. Let a vector $\vec{V}_1 = \overrightarrow{OP}$, where O is origin and P is (2, 0, 2). Vector $\vec{V}_1$ is rotated about origin by an angle of $\frac{\pi}{3}$ in a plan which perpendicular to XOZ plane to get vector $\vec{V}_2$ where $\vec{V}_2 \cdot \hat{j} > 0$. Again $\vec{V}_1$ is rotated about origin by right angle in XOZ plane to get vector $\vec{V}_3$. If V is volume of tetrahedron whose coterminous edges are $\vec{V}_1, \vec{V}_2$ and $\vec{V}_3$ then value of $\frac{3V}{\sqrt{6}}$ is _____.

Ans. 4

Sol. Equation of plane perpendicular to XOZ plane

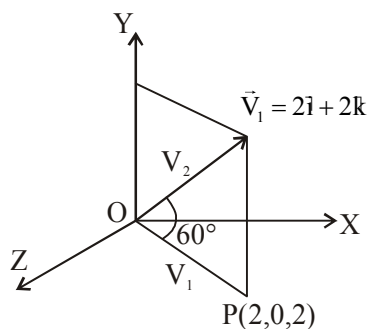
Is P : $x - z = 0$ $\vec{V}_2$ is coplanar with $\vec{V}_1$ and $\hat{j}$

$$\Rightarrow \vec{V}_2 = x\vec{V}_1 + y\hat{j}$$

$$\Rightarrow \vec{V}_2 = x(2\hat{i} + 2\hat{k}) + y\hat{j}$$

$$\therefore |\vec{V}_2| = |\vec{V}_1| = 2\sqrt{2}$$

$$\therefore 4x^2 + y^2 + 4x^2 = 8 \quad \text{_____ (1)}$$



$$\text{Now, } \vec{V}_2 \cdot \vec{V}_1 = |\vec{V}_2| |\vec{V}_1| \cos \frac{\pi}{3} = 4$$

$$4 = (x(2\hat{i} + 2\hat{k}) + y) \cdot (2\hat{i} + 2\hat{k})$$

$$4 = 8x \Rightarrow x = \frac{1}{2} \Rightarrow y = \pm\sqrt{6}$$

$$\Rightarrow \vec{V}_2 = \hat{i} + \sqrt{6}\hat{j} + \hat{k} \left(\because \vec{V}_2 \cdot \hat{j} > 0 \right)$$

$$\text{Now, } \vec{V}_3 = \vec{V}_1 + x\hat{i}$$

$$\because \vec{V}_3 \cdot \vec{V}_1 = 0 \Rightarrow 8 + 2x = 0 \Rightarrow x = -4$$

$$\Rightarrow \vec{V}_3 = 2\hat{i} + 2\hat{k} - 4\hat{i}$$

$$\vec{V}_3 = -2\hat{i} + 2\hat{k}$$

$$\Rightarrow \text{volume of tetrahedron } V = \left| \frac{1}{6} \begin{vmatrix} 2 & 0 & 2 \\ 1 & \sqrt{6} & 1 \\ -2 & 0 & 2 \end{vmatrix} \right|$$

$$\Rightarrow V = \frac{1}{6} 8\sqrt{6} \Rightarrow \frac{3V}{\sqrt{6}} = 4$$

V01/4/071. The volume of a tetrahedron OABC formed by the vectors $\overrightarrow{OA} = \vec{a}$, $\overrightarrow{OB} = \vec{b}$ and $\overrightarrow{OC} = \vec{c}$ where

$|\vec{a}| = 3$, $|\vec{b}| = 4$, $|\vec{c}| = 6$, angle between the vector pairs $\{\vec{a}, \vec{b}\}$, $\{\vec{b}, \vec{a}\}$ and $\{\vec{c}, \vec{a}\}$ be $\frac{\pi}{3}$, $\cos^{-1} \frac{1}{4}$ and

$\cos^{-1} \left(\frac{-1}{18} \right)$ respectively is V. If the value of V^2 can be written as $\frac{pq}{r^2}$ where p, q, r are prime numbers then

evaluate $\frac{p+q}{r}$ _____.

Ans. 30

Sol.
$$[\vec{a} \vec{b} \vec{c}]^2 = \begin{vmatrix} \vec{a} \cdot \vec{a} & \vec{a} \cdot \vec{b} & \vec{a} \cdot \vec{c} \\ \vec{b} \cdot \vec{a} & \vec{b} \cdot \vec{b} & \vec{b} \cdot \vec{c} \\ \vec{c} \cdot \vec{a} & \vec{c} \cdot \vec{b} & \vec{c} \cdot \vec{c} \end{vmatrix} = \begin{vmatrix} 9 & 6 & -1 \\ 6 & 16 & 6 \\ -1 & 6 & 36 \end{vmatrix}$$

$$= 9(576 - 36) - 6(216 + 6) - (36 + 16)$$

$$= 4860 - 1332 - 52 = 3476$$

$$\therefore \text{Volume of tetrahedron} = V = \frac{1}{6}[\vec{a} \vec{b} \vec{c}]$$

$$\therefore V^2 = \frac{1}{36} \cdot [\vec{a} \vec{b} \vec{c}]^2 = \frac{1}{36} \times 3476$$

$$= \frac{869}{9} = \frac{11 \times 79}{3^2} = \frac{pq}{r^2}$$

$$\therefore \frac{(p+q)}{r} = \frac{11+79}{3} = 30$$

V01/4/072. Three lines are given by $\vec{r} = \lambda \hat{i}, \lambda \in \mathbb{R}$, $\vec{r} = \mu(\hat{i} + \hat{j}), \mu \in \mathbb{R}$ and $\vec{r} = \nu(\hat{i} + \hat{j} + \hat{k}), \nu \in \mathbb{R}$. Let the lines cut the plane $x + y + z = 1$ at the points A, B and C respectively. If the volume of a tetrahedron OABC is V (Where O origin), then the value of $18V$ equals _____.

Ans. 0.5

Sol. A (1, 0, 0), B $\left(\frac{1}{2}, \frac{1}{2}, 0\right)$ and C $\left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right)$

$$\vec{OA} = \hat{i}, \vec{OB} = \frac{1}{2}\hat{i} + \frac{1}{2}\hat{j}$$

$$\vec{OC} = \frac{1}{3}\hat{i} + \frac{1}{3}\hat{j} + \frac{1}{3}\hat{k}$$

$$\Rightarrow V = \frac{1}{36}$$

V01/4/073. Let 'n' represents the number of Lattice points (x, y, z all being integers) inside the tetrahedron (not on the

surface) having vertices $(0, 0, 0)$, $(8, 0, 0)$, $(0, 8, 0)$ and $(0, 0, 8)$ then the value of $\left(\frac{n}{7}\right)$ is _____.

Ans. 5

Sol. $x + y + z < 8$, $x, y, z \geq 1$
 $\Rightarrow x + y + z + w = 8$
 $w \geq 1$
 $\Rightarrow$ No of solutions $= n = {}^7C_3 = 35$

V01/4/074. The equation of plane which passes through the line of intersection of the planes $x + y + z - 1$ and $x - y + z - 3 = 0$ and which is at maximum distance from the point $(3, 4, 5)$ is $ax + by + cz = d$ then the value of $\frac{a+b-c}{4d}$ _____.

Ans. 1.25

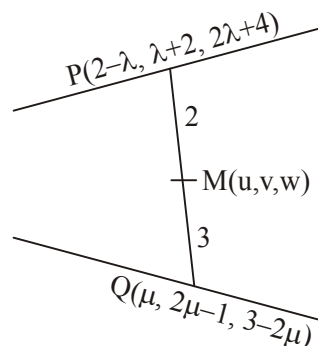
Sol. Any point on the line of intersection is $(\lambda + 2, -1, -\lambda)$ for maximum distance
 $= ((\lambda + 2 - 3)\hat{i} + (-1 + 4)\hat{j} + (-\lambda - 5)\hat{k}) \cdot (\hat{i} - \hat{k}) = 0$
 $\Rightarrow \lambda = -2$
 Normal $\vec{n} = 3\hat{i} + 5\hat{j} + 3\hat{k}$
 equation of plane will be $3x + 5y + 3z = 1$

V01/4/075. Consider two straight lines $L_1 : \frac{x-2}{-\frac{1}{2}} = \frac{y-2}{\frac{1}{2}} = \frac{z-4}{1}$ and $L_2 : \frac{x-0}{\frac{1}{2}} = \frac{y+1}{1} = \frac{z-3}{-1}$. Let P and Q be the variable on L_1 and L_2 respectively and point M divides PQ in the ratio 2 : 3 internally. If point M moves in a plane $ax + bz = 6$, then value of $a + b$ is _____.

Ans. 3

Sol. $L_1 : \frac{x-2}{-1} = \frac{y-2}{1} = \frac{z-4}{2}$ and $L_2 : \frac{x-0}{1} = \frac{y+1}{2} = \frac{z-3}{-2}$

Using section formula and eliminating



λ, μ we get $w + 2\mu = 6$

$M(u, v, w)$ moves on

$$2x + z = 6$$

V01/4/076. If $\vec{a}, \vec{b}, \vec{c}$ be three non-coplanar unit vectors such that the angle between every pair of them is 60° . If $p_i, q_i, r_i; i = 1, 2, 3$ are scalars satisfying $\vec{a} \times \vec{b} + \vec{b} \times \vec{c} = p_1\vec{a} + q_1\vec{b} + r_1\vec{c}$, $\vec{b} \times \vec{c} + \vec{c} \times \vec{a} = p_2\vec{a} + q_2\vec{b} + r_2\vec{c}$ and

$$\vec{c} \times \vec{a} + \vec{a} \times \vec{b} = p_3\vec{a} + q_3\vec{b} + r_3\vec{c} \text{ then } \begin{vmatrix} p_1 & q_1 & r_1 \\ p_2 & q_2 & r_2 \\ p_3 & q_3 & r_3 \end{vmatrix} = \lambda \begin{vmatrix} p_1 + q_1 & q_1 & r_1 \\ p_2 - r_2 & q_2 & r_2 \\ p_3 + 2q_3 & q_3 & r_3 \end{vmatrix}. \text{ The value of } \lambda \text{ is equal to } \underline{\hspace{2cm}}.$$

Ans. 1

Sol.

MATCH THE COLUMN TYPE QUESTION(S)

C01/5/077. Match the following column :

निम्न कॉलम का सुमेलित कीजिए :

Column-I

(a) In the triangle ABC, equation of the perpendicular bisector of the sides AB and AC are $x + y = 0$ and $y - x = 0$ respectively. If $A \equiv (5, 7)$ then equation of side BC is

(b) The straight line of the family $(x + y) + \lambda(2x - y + 1) = 0$ that is farthest from $(1, -3)$ is

(c) In triangle ABC, $A(1, 1)$, $B(4, -2)$, $C(5, 5)$. (r) $15y - 6x = 7$
Equation of the internal angle bisector of angle A is

(d) The diagonals AC and BD of a rhombus intersect at $(5, 6)$ If $A = (3, 2)$ then equation of diagonal BD is

Column-II

(p) $7y - 5x = 0$

(q) $y = 1$

(r) $15y - 6x = 7$

(s) $2y + x = 17$

Choose the correct option :

सही विकल्प का चयन कीजिए :

(A) $a - r, b - p, c - q, d - s$

(B) $a - p, b - r, c - q, d - s$

(C) $a - r, b - q, c - p, d - s$

(D) $a - r, b - q, c - s, d - p$

Ans. B

Sol.

C02/5/078. $A(-2, 0)$ and $B(2, 0)$ are two fixed points and P is a point such that $PA - PB = 2$. Let S be the circle $x^2 + y^2 = r^2$. Then match the following.

Match the following column :

निम्न कॉलम का सुमेलित कीजिए :

Column-I

Column-II

- (a) If $r = 2$, then the number of points P satisfying $PA - PB = 2$ and lying on $x^2 + y^2 = r^2$ is (p) 2
- (b) If $r = 1$, then the number of points satisfying $PA - PB = 2$ and lying on $x^2 + y^2 = r^2$ is (q) 4
- (c) For $r = 2$, the number of common tangents is (r) 0
- (d) For $r = \frac{1}{2}$, the number of common tangents is (s) 1
- (t) 5
- (u) 6

Choose the correct option :

सही विकल्प का चयन कीजिए :

- (A) $P \rightarrow 6$; $Q \rightarrow 4$; $R \rightarrow 3$; $S \rightarrow 5$
- (B) $P \rightarrow 2$; $Q \rightarrow 1$; $R \rightarrow 1$; $S \rightarrow 5$
- (C) $P \rightarrow 1$; $Q \rightarrow 4$; $R \rightarrow 3$; $S \rightarrow 1$
- (D) $P \rightarrow 2$; $Q \rightarrow 4$; $R \rightarrow 3$; $S \rightarrow 5$

Ans. C

Sol.

C02/5/079. Match the following column :

निम्न कॉलम का सुमेलित कीजिए :

Column-I

Column-II

- (a) Let $2x^2 + y^2 = 3xy$ be the equation of pair of tangents drawn from origin to circle of radius 3 with center in first quadrant. If O is origin and A is one of the points of contact, then $OA - 3\sqrt{10}$ is equal to. (p) 2
- (b) Let $2x + 3y = 1$ is tangent to a circle at $(1, -1)$. This circle is orthogonal to a circle which is drawn having diameter as a line segment with end points $(0, -1)$ and $(-2, 3)$. If $x^2 + y^2 + 2gx + 2fy + c = 0$ is the equation of circle, then $4f - 4g$ is equal to. (q) 3
- (c) If S and T are two concentric circles, such that from any point P on S pair of tangents drawn to T, touch it at A, B and centroid of $\triangle ABP$ lies on T. Then ratio of radii of S to T is (r) 5
- (d) If A, B lie on $x^2 + y^2 = 5$ such that tangents at A and B intersect at $P(5, -5)$, and (α, β) is ortho-center of $\triangle ABP$, then $\alpha + \beta + 3$ is equal to (s) 9

Choose the correct option :

सही विकल्प का चयन कीजिए :

- (A) a-r, b-s, c-p, d-q (B) a-r, b-p, c-s, d-q (C) a-s, b-r, c-p, d-q (D) a-r, b-p, c-q, d-s

Ans. C

Sol.

C04/5/080. Let $E \equiv \frac{x^2}{4} + \frac{y^2}{3} = 1$, $H(\alpha, 0)$, $0 < \alpha < 2$. A straight line L drawn through H and parallel to Y -axis crosses the ellipse E and its auxiliary circle at points E_E and E_A respectively in the first quadrant. The tangent to ellipse E at E_E intersects the positive X -axis at point G and the line joining E_A and origin makes an angle θ with positive X -axis.

Match the following column :

निम्न कॉलम का सुमेलित कीजिए :

Column-I (Value of θ)

Column-II (Area of $\Delta E_A GH$)

(a) $\frac{\pi}{4}$

(p) $\frac{(\sqrt{3}-1)^4}{8}$

(b) $\frac{\pi}{3}$

(q) 1

(c) $\frac{\pi}{6}$

(r) $\frac{3}{4}$

(d) $\frac{\pi}{12}$

(s) $\frac{1}{2\sqrt{3}}$

(t) $\frac{3\sqrt{3}}{2}$

Choose the correct option :

सही विकल्प का चयन कीजिए :

(A) $a - t$, $b - q$, $c - s$, $d - p$

(B) $a - q$, $b - t$, $c - s$, $d - p$

(C) $a - t$, $b - s$, $c - q$, $d - p$

(D) $a - t$, $b - s$, $c - p$, $d - q$

Ans. B

Sol.

C05/5/081. Consider four conics defined as follows

$$C_1 : \frac{x^2}{10} - \frac{y^2}{4} = 1; C_2 : \frac{x^2}{3} + \frac{y^2}{4} = 1; C_3 : \frac{y^2}{2} - x^2 = 1; C_4 : x^2 = 16y$$

Match the following column :

निम्न कॉलम का सुमेलित कीजिए :

Column-I

Column-II

(a) One of the tangent to C_1 is

(p) $y = 2x + 9$

(b) One of the tangent to C_2 is

(q) $y = 2x + 7$

(c) One of the normal to C_3 is

(r) $y = 2x + 4$

(d) One of the normal to C_4 is

(s) $y = 2x + \frac{3}{\sqrt{5}}$

(t) $y = 2x + 6$

(u) $y = 2x + \frac{6}{\sqrt{7}}$

Choose the correct option :

सही विकल्प का चयन कीजिए :

(A) $P \rightarrow t; Q \rightarrow r; R \rightarrow u : S \rightarrow p$

(B) $P \rightarrow q; Q \rightarrow s; R \rightarrow p : S \rightarrow r$

(C) $P \rightarrow t; Q \rightarrow r; R \rightarrow s : S \rightarrow p$

(D) $P \rightarrow t; Q \rightarrow p; R \rightarrow u : S \rightarrow r$

Ans.

A

Sol.

Paragraph for Question C03/3/082 to C03/3/084

Column -I : contains conics; Column-2 : number of common tangents; Column-3 : number of common Normal

Column - 1	Column - 2	Column - 3
(I) $2y^2 = 2x - 1$ and $2x^2 = 2y - 1$	(i) 0	(P) 0
(II) $(y-1)^2 = \frac{4}{\sqrt{3}} \left(x + \frac{\sqrt{3}}{4} \right)$ and $\frac{\left(x + \frac{\sqrt{3}}{4} \right)^2}{9} + \frac{(y-1)^2}{3} = 1$	(ii) 1	(Q) 1
(III) $x^2 = y - 2$ and $y + x^2 - 2x + 3 = 0$	(iii) 2	(R) 2
(IV) $y^2 = -4x$ and $x^2 = -4y$	(iii) 3	(S) 3

C03/3/082. Which of the following options is the only CORRENT combination :

(A) (I) (iv) (Q)

(B) (I) (iv) (S)

(C) (II) (i) (S)

(D) (II) (ii) (S)

Ans.

A

Sol.

C03/3/083. Which of the following options is the CORRECT combination :

(A) (IV) (i) (Q)

(B) (IV) (i) (P)

(C) (IV) (ii) (Q)

(D) (I) (iv) (P)

Ans.

C

Sol.

C03/3/084. Which of the following options is the only INCORRECT combination :

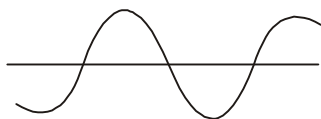
- (A) (III) (iii) (Q) (B) (II) (i) (S) (C) (II) (iii) (S) (D) (IV) (ii) (Q)

Ans. B

Sol. (I) $2y^2 = 2x - 1$ and $2x^2 = 2y - 1$

Equation of tangents are $y = m\left(x - \frac{1}{2}\right) + \frac{1}{4m}$ and $y - \frac{1}{2} = mx - \frac{m^2}{4}$ respectively

$$\text{So, } -\frac{m}{2} + \frac{1}{4m} = \frac{1}{2} - \frac{m^2}{4}; m^3 - 2m^2 - 2m + 1 = 0$$



This equation has 3 real and distinct roots i.e., 3 common tangents. For common normals, they are symmetric about $y = x$

So, only one common normal

(II) After shifting at origin, we get $y^2 = \frac{4}{\sqrt{3}}x$ and $\frac{x^2}{9} + \frac{y^2}{3} = 1$

Equation of normals are $y = mx - \frac{2}{\sqrt{3}}m - \frac{1}{\sqrt{3}}m^3$ and $y = mx \pm \frac{m(a^2 - b^2)}{\sqrt{a^2 + b^2m^2}}$

$$\text{So, } \frac{2}{\sqrt{3}}m + \frac{1}{\sqrt{3}}m^3 = \frac{6m}{\sqrt{9 + 3m^2}}; m = 0, \pm 1 \text{ Hence, 3 common normals}$$

(III) $y = x^2$ and $y = -(x - 1)^2$

Equation of tangents are $y = mx - \frac{m^2}{4}$ and $y = m(x - 1) + \frac{m^2}{4}$ respectively

So, we get $m = 0, 2$. Hence, 2 common tangents

Common normals : Equation are $y = mx + \frac{1}{2} + \frac{1}{4m^2}$ and $y = mx - m - \frac{1}{4m^2} - \frac{1}{2}$

$$\text{So, we get } 2m^3 + 2m^2 + 1 = 0.$$

Only one real root, so 1 common normal.

(IV) $y^2 = -4x$ and $x^2 = -4y$

Number of common tangents will be same

For $y^2 = -4x$ and $x^2 = -4y$, only one common tangent

$$\text{Common normals : } 2m + m^3 = -2 - \frac{1}{m^2}$$

$f(m) = m^5 + 2m^3 + 2m^3 + 1$, only one root negative, So, only one common normal.